

Introduction To Statistical Mechanics

1. Introduction

Our discussion will be of the systems consisting of large number of particles. Examples: gases, liquids, solids, electromagnetic radiation (photons), etc. Since most physical, chemical, or biological systems consist of large number of molecules; so our discussion is relevant to the large part of universe.

System consisting of many particles can be divided into two parts:

(a) **High-energy physics:** The challenge is to understand the *fundamental interactions* between nucleons, neutrinos, mesons, or other strange particles.

(b) **Many particles Systems:** To discuss solids, liquids, plasmas, chemical or biological systems, and other such systems involving many particles this is right now our area of interest.

Although it might be possible to write down the *equations of the motion* for any one of these systems, the complexity of a system containing many particles is so great that it may make the

task of getting any useful results or predictions almost hopeless or impossible. It may require very deep analysis to predict the occurrence of these features from knowledge of the individual particles.

For example, it is a striking fact, and one which is difficult to understand in microscopic detail, that simple atoms forming a gas can condense abruptly to form a liquid with very different properties. It is a fantastically more difficult task to attain an understanding of how an assembly of certain kinds of molecules can lead to a system capable of biological growth and reproduction.

Knowing Interactions: The task of understanding systems consisting of many particles is thus far from trivial, even when the interactions between individual atoms are well known. The main aim is, instead:

- (a) Knowledge of basic physical laws is used to develop new concepts that can illuminate the essential characteristics of such complex systems.
- (b) Provide sufficient insight to facilities one's thinking, to recognise important relationships, and to make useful predictions.

Sizes of Systems: It is useful to introduce a distinction between the sizes of systems whose description may be of interest.

(a) **Microscopic:** We shall call a system "microscopic" (i.e., "small scale") if it is roughly of atomic dimensions or smaller (say of the order of 10^{-10} Å or less). For example, the system might be a molecule.

(b) **Macroscopic:** On the other hand, we shall call a system "macroscopic" (i.e., "large scale") when it is large enough to be visible in the ordinary sense (say greater than 1 micron, so that it can at least be observed with a microscope using ordinary light). The system consists then of very many atoms or molecules. For example, it might be a solid or liquid of the type we encounter in our daily experience.

Macroscopic Parameters: Quantities like volume, pressure, magnetic moment (There are two magnetic polarities, each of which is used to represent either 0 or 1.), thermal conductivity, etc. (Temperature, heat, thermal energy, and entropy are all what we call **macroscopic measurements**)

Equilibrium: If the macroscopic parameters of an isolated system do not vary in time, then one says that the system is in equilibrium.

The boundaries between these disciplines (Thermodynamics and Statistical mechanics) are not very sharp; it may be useful to point out briefly the similarities and differences between their respective methods of approach.

(a) **Classical Thermodynamics:** For a system in equilibrium, one can try to make some very general statements concerning relationships existing between the macroscopic parameters of the system. This is the approach of classical “thermodynamics, historically the oldest discipline. The strength of this method is its great generality, which allows it to make valid statements based on a minimum number of postulates without requiring any detailed assumptions about the microscopic (i.e., molecular) properties of the system. The strength of the method also implies its weakness: only relatively few statements can be made on such general grounds, and many interesting properties of the system remain outside the scope of the method.

(b) **Statistical Mechanics:** For a system in equilibrium, one can again try to make very general statements consistently based, however, on the microscopic properties of the particles in the system and on the laws of mechanics governing their behaviour. This is the approach of “statistical mechanics.” It yields all the results of thermodynamics plus a large number of general relations for calculating the macroscopic parameters of the system from knowledge of its microscopic constituents. This method is one of great beauty and power.

(c) **Irreversible Thermodynamics: or statistical mechanics of irreversible processes:** If the system is not in equilibrium, one faces a much more difficult task. One can still attempt to make very general statements about such systems, and this leads to the methods of “irreversible thermodynamics,” or, more generally, to the study of “statistical mechanics of irreversible processes.” But generally and power of these methods is much more limited than in the case of systems in equilibrium.

(d) One can attempt to study in detail the interactions of all the particles in the system and thus to calculate parameters of macroscopic significance. This is the method of “kinetic theory.” It is in principle always applicable, even when the system is not in equilibrium so that the powerful methods of equilibrium statistical mechanics are not available. Although kinetic theory yields the most detailed description, it is by the same token also the most difficult method to apply.

Furthermore, its detailed point of view may tend to obscure general relationships which are of wider applicability.

Historically: The subject of thermodynamics arose first before the atomic nature of matter was understood.

(a) The idea that heat is a form of energy was first suggested by the work of Count Rumford (1798) and Davy (1799).

(b) It was stated explicitly by the German physician R. J. Mayer (1842) but gained acceptance only after the careful experimental work of Joule (1843 - 1849).

(c) The first analysis of heat engines was given by the French engineer S. Carnot in 1824.

(d) Thermodynamic theory was formulated in consistent form by Clausius and Lord Kelvin around 1850, and was greatly developed by J. W. Gibbs in some fundamental papers (1876 - 1878).

(e) The atomic approach to macroscopic problems began with the study of the kinetic theory of dilute gases. This subject was developed through the pioneering work of Clausius, Maxwell, and Boltzmann. Maxwell discovered the distribution law of molecular velocities in 1859, while Boltzmann formulated his fundamental integrodifferential equation (the Boltzmann equation) in 1872.

(f) The kinetic theory of gases achieved its modern form when Chapman and Enskog (1916 - 1917) succeeded in approaching the subject by developing systematic methods for solving this equation.

(g) The more general discipline of statistical mechanics also grew out of the work of Boltzmann who, in 1872, succeeded further in giving a fundamental microscopic analysis of irreversibility and of the approach to equilibrium.

(h) The theory of statistical mechanics was then developed greatly in generality and power by the basic contributions of J. W. Gibbs (1902). Although the advent of quantum mechanics has brought many changes, the basic framework of the modern theory is still the one which he formulated.

This may seem rather surprising at first sight; for are not systems such as gases or liquids, which consist of a number of particles of the order of Avogadro's number ($\sim 10^{23}$), hopelessly

complicated? The answer is that the *very complexity of these systems contains within it the key to a successful method of attack*. It becomes possible to apply statistical arguments to them.

Some definitions:

- **Micro State:** A microstate is a specific microscopic configuration of a thermodynamic system that occupy with a certain property in the course of its thermal fluctuation. The position (x) momentum (p) energy (E), and spin (s, s_z) of individual atom are the example of microstate of system.
- **Macro State:** A macro state refers to macroscopic properties of system such as temperature (T), pressure (P), free energy (F) entropy (S). a macro state is characterize by a probability distribution of possible state across a certain statistical ensemble of all microstates, and distribution describes the probability of finding the system in certain microstate .
- **Accessible State:** Any microstate in which a system can be found without contradicting the macroscopic information available about the system.
- **Statistical Ensemble:** an assembly of large number of mutually non interacting systems, each of which satisfies the same conditions as those known to be satisfied by a particular system under condition. There are three types of ensemble (a) **micro canonical ensemble**, (b) **canonical ensemble**, (c) **Grand canonical ensemble**. An ensemble is said to be time independent ensemble if number of system exhibiting any particular property is the same at a time.
- **Probability:** The probability p_r of occurrence of an event r in a system is defined with respect to statistical ensemble of N such a systems. If N_r systems in the ensemble exhibit the event r then

$$p_r = \frac{N_r}{N}$$

- **Probability Density:** The probability density $\rho(u)$ is defined by the property that $\rho(u)du$ yields the probability of finding the continuous variable u in the range between u and $u + du$.

- **Mean Value:** The mean value of u is denoted by $\langle u \rangle$ as defined as $\langle u \rangle = \sum_r p_r u_r$ where the sum is over all possible value values u_r of the variable u and p_r is denotes the probability of occurrence of the particular value u_r . Above definition is for discrete variable .

For continuous variable u , $\langle u \rangle = \int u \rho(u) du$

- **Dispersions (or variance):** The dispersion of u is defined as $\sigma^2 = \langle (\Delta u)^2 \rangle$
 $= \sum_r p_r (u_r - \langle u \rangle)^2$ which is equivalent to $\sigma^2 = \langle (\Delta u)^2 \rangle = \sum_r \langle u^2 \rangle - \langle u \rangle^2$
- **Stirling Formula:** Stirling's approximation (or Stirling's formula) is an approximation for large factorials. It is named after James Stirling.

The formula as typically used in applications is $\ln N = N \ln N - N$